

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2, BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10

Total Marks:50

Annexure A

Constraint - Based Problem-Solving

Duration: 2Hrs

1. Develop a Python-based Reinforcement Learning model using the OpenAI Gym environment to solve a Grid World navigation problem where the agent must reach the goal while avoiding obstacles. Explain how the state space, action space, reward function, and constraints are defined.
(10 Marks)
2. Implement an ϵ -greedy Multi-Armed Bandit algorithm in Python to maximize rewards under a limited budget of 500 trials. Compare the effect of different ϵ values (0.1, 0.3, and 0.5) on exploration, exploitation, and constraint satisfaction.
(10 Marks)
3. Design a Markov Decision Process (MDP) for an autonomous robot that must reach a destination while minimizing energy consumption. Define the states, actions, transition probabilities, rewards, and energy constraints.
(10 Marks)
4. Using Python and TensorFlow/Keras, implement a value iteration algorithm to determine the optimal policy for a constrained grid environment where certain states are restricted. Explain how Bellman equations are used to obtain the optimal policy.
(10 Marks)

5. A warehouse robot has a battery capacity of only 30 minutes. Design an RL framework that enables the robot to complete the maximum number of deliveries without exceeding the battery limit. Explain the constraints, policy, and reward design.

(10 Marks)

6. Implement a Reinforcement Learning agent that controls traffic signals at an intersection while ensuring that emergency vehicles receive immediate priority. Explain how exploration, exploitation, and policy learning are modified to satisfy the safety constraint.

(10 Marks)

7. Develop a Python program using Keras/TensorFlow to simulate a reinforcement learning agent for packet routing in a wireless network with limited bandwidth. Explain how bandwidth constraints influence the reward function and policy optimization.

(10 Marks)

8. Analyse an experimental comparison between random action selection and ϵ -greedy action selection in a Multi-Armed Bandit problem under a fixed time constraint. Record cumulative rewards and explain which strategy provides better performance.

(10 Marks)

9. Design a Reinforcement Learning framework for a delivery drone that must maximize successful deliveries while avoiding no-fly zones and maintaining battery limits. Explain the MDP formulation, Bellman equation, and policy learning process.

(10 Marks)

10. Implement a Reinforcement Learning solution using Python for a smart energy management system where electricity consumption must remain below a specified threshold. Explain how reward shaping, policy learning, and feasibility constraints improve decision-making.

(10 Marks)

Code of Conduct:

I __Gk.vyshnavi_/192425069__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

vyshnavi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Develop a Reinforcement learning model for Grid world navigation problem.

problem understanding:

The objective is to design a reinforcement learning model in which an agent learns to navigate through a Grid world environment. Agents starts from an initial position and must reach a predefined goal while avoid obstacles and minimizing unnecessary movements. The agent improves its behaviour by interacting with the environment, and learning from rewards and penalties.

Constraints:

1. agent cannot move outside the grid boundaries.
2. obstacles cannot be crossed.
3. Goal should be reached in the minimum possible no. of steps.
4. episode ends when the goal is reached or the maximum no. of allowed steps is exceeded.

state space:

State space consist of all possible positions of the agent in the grid. Each cell in the grid represents one unique state. For example, in 5×5 grid.

Action space:

1. move up
2. move down
3. move left
4. move Right

Reward function:

Situation	Reward
Reaching the goal	+100
Moving to a valid cell	-1
Hitting an obstacle	-20
Exceeding maximum steps	-50

Constraint analysis:

The constraints ensure safe and efficient navigation.

1. Grid boundaries prevent illegal movements.
2. obstacles force the agent to discover alternative paths.
3. step penalties encourage shorter routes.
4. maximum step limits prevent infinite wandering.

2. ϵ -Greedy multi Armed Bandit Algorithm.

Problem understanding.

Multi armed Bandit problem involves selecting the best option from multiple alternatives to maximize total rewards.

Constraint:

1. maximum number of trials
2. Rewards should be maximized within the limited trial budget.

ϵ -greedy strategy:

1. with probability ϵ , the agent explores by selecting a random option.
2. with probability $1 - \epsilon$, the agent exploits by selecting the option with a highest estimated reward.

Constraint analysis:

$\epsilon = 0.1$

1. explores only occasionally.
2. quickly exploits the best performing option
3. Best suited for a limited trial budget.

$\epsilon = 0.3$

1. maintains a balance b/w learning and reward collection.
2. suitable when reward distributions are uncertain.

$\epsilon = 0.5$

1. frequently explores new options.
2. learns about all alternatives but spends fewer.

3. Markov Decision process (MDP) for an autonomous Robot

problem understanding:

an autonomous robot must reach its destination while minimizing energy consumption. The robot operates in a uncertain environment where actions may not always produce.

States:

1. current location.
2. Remaining battery level.
3. presence of nearby obstacles.

Actions:

1. move forward.
2. Turn left
3. Turn right
4. Recharge battery
5. Stop.

Transition probabilities:

Eg:

1. moving forward successfully: 90%.
2. Encountering an obstacle: 10%.

Constraint analysis:

1. Reach the destination quickly.
2. minimize energy consumption.

Value iteration for a constrained Grid environment problem understanding.

The objective is to determine the optimal policy for navigating a constrained grid where certain cells are restricted and cannot be entered.

Constraints:

1. Restricted states cannot be visited.
2. The goal state must be reached.
3. movement cost should be minimized.
4. Boundary conditions must be respected.

Value iteration:

value iteration is a dynamic programming algorithm that computes the optimal value for every state.

Bellman Equation:

The Bellman equation states that the value of a state equals the immediate reward plus the discounted value of the best possible future state.

constraint analysis:

Restricted states are excluded from policy updates, ensures that the final policy never selects invalid paths. step costs encourage shorter and more efficient routes.

5. Reinforcement learning framework for a warehouse Robot

Problem understanding

A warehouse robot has a battery capacity of only 30 minutes. The objective is to maximize the number of deliveries while ensuring that the battery never becomes exhausted.

Constraints:

1. maximum battery life is 30 min
2. each movement consumes battery power
3. The robot must recharge before the battery is depleted.
4. Battery capacity cannot be exceeded after recharging.

State space:

1. current robot location
2. Remaining battery level
3. number of completed deliveries.

Action space:

1. pick up a package.
2. Deliver a package
3. move to another location.
4. Return to the charging station.
5. Recharge the battery.